

Programme	B. Sc. Mathematics Honours				
Course Code	MAT6FS113 (2)				
Course Title	Scientific Principles & Practice				
Type of Course	SEC (for pathways 1 – 5)				
Semester	VI				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	3	3	-	-	45
Pre-requisites	High School science				
Course Summary	This course familiarises students with the basic principles and phenomenology of science and scientific research.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the scope, limitations, and fundamental principles of science and scientific research.	U	C	Seminar Presentation/ Group Tutorials
CO2	Appreciate the role of abstraction and critical thinking in mathematics and science, and how they contribute to scientific progress.	U	M	Seminar Presentation/ Group Tutorials
CO3	Recognize the importance of proper experimental design in conducting effective scientific research.	U	C	Seminar Presentation/ Group Tutorials
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text Book	The Scientific Endeavour – A Primer on Scientific Principle & Practice, 2 nd Edition, Jeffrey A. Lee (2016).			
Module	Unit	Content	Hrs (36 +9)	Marks (50)
I	The Philosophy of Science		9	Min10
	Chapter 1 - Introduction			
	1	1.1: What is Science?		
	2	1.2: Areas of Science		
	3	1.3: Basic & Applied Research		
	4	1.4: Why Understand Science?		
	Chapter 2 - The Philosophy of Science			
	5	2.1: Scientific Statements		
	6	2.2: Scientific Methods		
	7	2.3: Recent Development in the Philosophy of Science		
II	Scientific Research		9	Min10
	Chapter 3 – Research			
	8	3.1, 3.2: Selecting a Topic, Hypothesis		
	9	3.3: Experimental Design		
	10	3.4: Performing Experiments		
	11	3.5-3.8: Analysis, Results, Discussion, Models		
	12	3.9: Non-experimental Research		
	Chapter 4 – The Community of Scientists			
	13	4.1: Scientific Norms		
	14	4.2-4.5: Invisible Colleges, Peer Review, Reward System, Becoming a Scientist		
III	Misconduct in Science & Critical Thinking		9	Min10
	Chapter 5 – Misconduct in Science			
	15	5.1: Fraud		
	16	5.2: Plagiarism		
	17	5.3: Questionable Research Practices		
	18	5.4: Research With Human & Animal Subjects		
	19	5.5: Whistleblowing		
	Chapter 6 – Critical Thinking & Science			
	20	6.1: Critical Thinking Strategies		
	21	6.2: Common Fallacies		
IV	Pseudoscience		9	Min10
	22	Chapter 7: 7.1-7.9: - Common Pseudosciences		
	23	8.1: Science & Pseudoscience		
	24	8.2: The Need for Critical Thinking		
	25	8.3: A Sceptical Attitude		
	26	8.4: Evaluating Extraordinary Claims		
	27	9.1: The Scientific Knowledge Acquisition Web		
	28	9.2: Conclusions		
V	Open Ended Module		9	
	1	Flatland: A Romance of Many Dimensions, Edwin Abbott Abbott, 1884.		

	2	Mr. Tompkins in Paperback, George Gamow, Cambridge University Press, 1993.		
	3	The Character of Physical Law, Richard Feynman, MIT Press, 2017.		
References:				
1. Mathematics & The Laws of Nature, John Tabak. 2. The Scientific Method: A Historical & philosophical Introduction, Barry Gower 3. History & philosophy of Science: A Reader, Daniel J. McKaughan & Holly VandeWall 4. A Historical Introduction to the Philosophy of Science, 4 th Edition, John Losee 5. A Summary of Scientific Method, Peter Kosso 6. The Nature of Physical Reality, Henry Margenau				

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	2	1	2	2	3	2	3	2	3	2	3
CO 2	3	2	2	3	3	2	2	2	3	2	3
CO 3	2	1	3	2	3	2	3	2	3	2	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓

VALUE-ADDED COURSES
(VAC)